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Decomposition–Ensemble Approach for Realized Volatility Prediction

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(IMFs) and residuals, effectively capturing the intricate patterns inherent in the data. Sample entropy and t-tests are utilized to classify these components into high-frequency and low-frequency segments. The low-frequency components, representing long-term trends, are directly fed into various Deep Neural Network (DNN) architectures, including Simple Recurrent Neural Network (SRNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional GRU (BiGRU), and Bidirectional LSTM (BiLSTM). High-frequency components, capturing short-term fluctuations, are modeled using Generalized Autoregressive Conditional Heteroskedasticity (GARCH) to extract conditional volatility, which is subsequently inputted into the DNN models. The final realized volatility prediction is obtained by linearly superimposing the outputs from all components, forming a comprehensive ensemble model. The proposed hybrid models are benchmarked against traditional standalone DNN models using evaluation metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). Empirical results demonstrate that the hybrid VMD-GARCH-DNN models

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Data Availability

Data available on request and the codes it will be available to public once the manuscript has

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